

# GEORGE SAAD

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## Experience

### Amazon | Applied Scientist II

New York City, NY | Mar. 2025 – Present

- Built and productionized a **large-scale RAG-based, agentic LLM system**, emphasizing scalable alignment methods (human-in-the-loop prompting, generate-and-rank) deployed across 50M+ products and 100M+ daily searches
- Designed **post-training preference alignment pipelines using SFT** and custom loss functions to optimize generation for purchase intent, relevance & brand sentiment, directly driving \$XXX M+ in annualized revenue
- Engineered an **RLAIF** loop using **LLM-as-a-judge** to generate preference data at 1M+ pairs/week and **DPO-tuned** guardrail models against quality and brand-safety regressions; driving a 50% increase in ad CTR
- Authored **2 research papers** and **1 patent submission** on scalable LLM alignment, evaluation, and controllable generation for advertising systems

### University of Toronto | Research Assistant

Toronto, ON | May 2023 – Feb. 2025

- Conducted research on **LLM-integrated conversational recommender systems**, combining retrieval, query reformulation, and generation to improve recommendation quality
- Led projects resulting in **peer-reviewed publications at ACL, SIGIR, and RecSys**, focusing on LLM-based retrieval, reasoning, and evaluation

### Amazon | Applied Scientist Intern

Seattle, WA | Jun. 2024 – Sep. 2024

- Developed an **agentic, multi-stage LLM-based semantic disambiguation pipeline** that filtered data early and applied deeper LLM reasoning only to ambiguous cases, maintaining inference efficiency while improving product search ranking by 29% across 8.5M+ weekly queries
- Evaluated failure modes of **embedding-only semantic methods** and identified ambiguity regimes where LLM-based reasoning materially improved clustering and classification quality
- Implemented **production-ready modeling, evaluation, and logging code**, enabling integration with Amazon's search stack and reliable offline/online validation

### Vector Institute (Geoffrey Hinton) | Applied Machine Learning Scientist

Toronto, ON | Jan. 2023 – Sep. 2023

- Evaluated deep causal inference models (**TARNet, DragonNet**) on observational datasets, studying sensitivity to confounding and lack of ground-truth counterfactuals
- Implemented evaluation pipelines to compare causal estimators on synthetic & real-world data under distributional shift

### Meta (Instagram) | Software Engineer Intern

Menlo Park, CA | May 2022 – Jul. 2022

- Built a **high-throughput Thrift cross-language service** to track 260M+ post-click external interactions daily, enabling richer feedback signals for personalized ad delivery and ranking on Instagram
- Developed app-level detection logic (Swift/Kotlin) to surface cross-app signals used for more accurate ad targeting, expanding delivery scope and **increasing monetizable impressions by 120%** for previously unlinked users

### FoodTrain | Founding Engineer

Sunnyvale, CA | Mar. 2020 – Dec. 2020

- Directed a 10-person engineering team across product design, system architecture, and execution, shipping a full-stack food delivery MVP (**React Native, Node.js, MongoDB**) from 0 to 1 in under 6 months
- Owned market research and product strategy — built a **Python** data pipeline to scrape and analyze 900+ competitor data points, translating insights into product roadmap and UX decisions

## Publications / Research

### Q-STRUM Debate: Query-Driven Contrastive Summarization for Recommendation Comparison | Paper

G. Saad, S. Sanner - *Findings of ACL 2025*

### Elaborative Subtopic Query Reformulation for Broad and Indirect Queries in Travel Dest. Rec. | Paper

Q. Wen\*, Y. Liu\*, J. Zhang\*, G. Saad, A. Korikov, Y. Sambale, S. Sanner - *ROEGEN @ RecSys 2024*

### Multi-Aspect Reviewed-Item Retrieval via LLM Query Decomposition and Aspect Fusion | Paper

A. Korikov\*, G. Saad\*, E. Baron, M. Khan, M. Shah, S. Sanner - *IR-RAG @ SIGIR'24*

## Education

### University of Toronto

Sep. 2023 – Jan. 2025

- MASc (Thesis-based), Research in LLMs & Recommendation supervised by Prof. Scott Sanner, GPA: 4.0 (out of 4.0)

### University of Toronto

Sep. 2019 – May 2023

- BASc in Engineering Science, Major in Machine Intelligence, Certificate in Engineering Business, GPA: 3.76 (out of 4.0)

## Technical Skills

**Languages:** Python, JavaScript, Java, C/C++, SQL

**Technologies:** PyTorch, JAX, vLLM, HuggingFace, Scikit-learn, Numpy, Pandas, AWS (EC2, S3, SageMaker, Bedrock), Apache Spark, Docker, Git, React, Node.js, Flask, Django, MongoDB